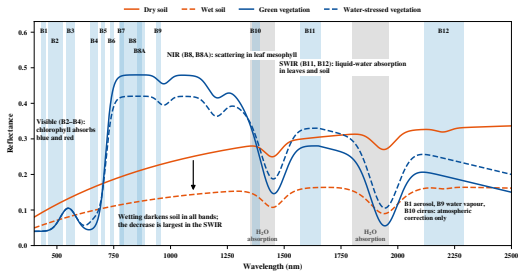
Motivation Different ways of obtaining soil moisture

Source	How SM is obtained	Depth	Main limitation
Microwave (SMAP, SMOS, ASCAT, S1)	invert: σ^0 or $T_B = f(\text{SM}, \text{roughness}, \text{vegetation})$	top $\sim$5 cm; informs part of the root zone (Feldman et al., 2023)	shallow; assumes vegetation and roughness; coarse (9–40 km) except S1
Optical / thermal (S2, Landsat, MODIS)	empirical relations with surface properties: reflectance and indices; LST–vegetation space (e.g. triangle method, TVDI)	skin / surface	indirect, site-specific calibration; clouds; canopy hides the soil
Land-surface models	$d\theta/dt = P - ET - Q - D$, with parameters p	full profile	forcing errors, soil and vegetation parameters, process representation (Koster et al., 2009; Seneviratne et al., 2010)
Reanalysis (ERA5-Land)	LSM driven by reanalysis meteorology	full profile	model assumptions dominate deep layers with few observational constraints (Li et al., 2021; Miguez-Macho and Fan, 2021)
In situ (ISMN)	direct probe measurement	several depths	point scale; sparse, uneven global coverage
Data-driven (ML / DL)	$\text{SM} = f_\theta(\text{inputs})$, learned from in-situ labels; no ad-hoc process assumptions	any depth with labels	needs enough, representative sites; generalisation?; hard to interpret?

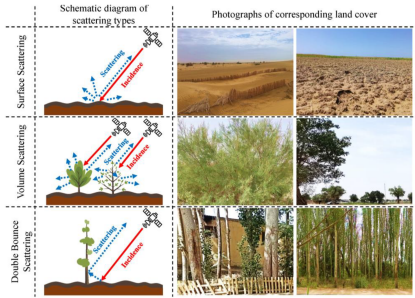
Motivation Multimodal remote sensing

1. Optical: reflectance spectrum



- Dry and wet soil reflect differently: wetting darkens soil in every band, most in the SWIR (B11, B12)
- Green and water-stressed vegetation reflect differently: stress lowers NIR and raises SWIR

2. SAR: scattering mechanisms



Nurmemet et al. (2025), *Sensors* 25(8), 2512, doi:10.3390/s25082512

- Bare soil: surface scattering, set by moisture (dielectric) and roughness
- Canopy: volume and double-bounce scattering mask the soil signal

Motivation Geospatial foundation models (GFMs)

What they are

- Large networks (usually Vision Transformers) pre-trained on huge *unlabelled* Earth-observation archives
- Self-supervised: learn by reconstructing or translating between sensors, no soil-moisture labels needed
- Output: *embeddings* — compact vectors that summarise what a patch of land looks like

Why for soil moisture

- In-situ labels are scarce ($\sim$1000 stations); pre-training brings knowledge from billions of pixels
- One shared representation for S2, S1, DEM and land cover

TerraMind (used here)

- Multi-modal, any-to-any GFM (IBM / ESA)
- ViT-Base: 12 layers, 768-dim, used *frozen*
- Input tile 2.24 km $\rightarrow$ $14 \times 14 = 196$ tokens, each a 160 m patch
- Modalities used: S2 L2A, S1 RTC, DEM, LULC
- Layers L3, L6, L9, L12 extracted

Open question

- Do these general-purpose embeddings carry soil-moisture information — or only describe the landscape?